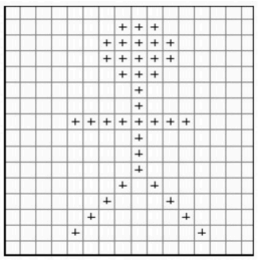
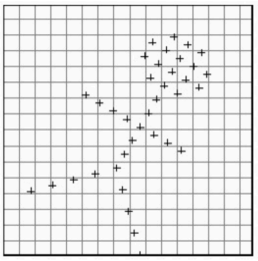
Bilinear_Interpolation

1. Motivation: When an image undergoes an affine transformation such as a rotation or scaling, the pixels in the image get moved around. Imagine a grayscale image (for simplicity) as a grid with the center of each grid having a float value. Most often the transformed coordinates (meshgrid) would be have decimal values and would not a corresponding pixel value in the original image.
2.  
3. So for example, suppose that after rotating an image, we need to find the pixel value at the location (6.7, 3.2). The problem with this is that there is no such thing as fractional pixel locations.
4. Image processing affine transformation usually follows the 3-step pipeline:
    1. Grid generator: first we create a sampling grid composed of (x, y) coordinates. For example, given a 400×400 grayscale image, we create a meshgrid of same dimension, that is, evenly spaced $x \in [0, W]$ and $y \in [0, H]$.
    2. We then apply the transformation matrix to the sampling grid generated in the step above.
    3. Grid sampler/interpolator (**bilinear interpolation**, etc): finally, we sample the resulting grid from the original image using the desired interpolation technique.
    4. In summary, to apply affine transformation to an image we do not apply the transformation to the image directly i.e. an apply transformation to an image of shape (H, W, C) instead, what we do is create a tensor containing image coordinates (meshgrid) of shape (H, W, 2). Then we apply transformation to the meshgrid coordinates. Finally, perform sampling or interpolation from the original image to get the pixel value for each grid cell in the transformed meshgrid. For example in grid cell with index x=2, y=3 we have (6.7, 2.3) is telling we have to sample from the coordinate x=6.7 and y=2.3 in the original image.
    5. Note: we must make sure that no value goes beyond the original image boundaries (use np.clip).
    6. Related: Spatial Transformer Networks.
5. Code:

```
# -*- coding: utf-8 -*-
"""
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@author: rslim
"""

import cv2
import math
import numpy as np
import torch
import torch.nn.functional as F

def linear_resize(in_array, size):
    """
    Parameters
    ----------
    in_array : input 1D array.
    size : desired size

    Returns
    -------
    resized array

    """
    ratio = (len(in_array) - 1) / (size - 1)
    out_array = []

    for i in range(size):
        low = math.floor(ratio * i)
        high = math.ceil(ratio * i)
        weight = ratio * i - low

        a = in_array[low]
        b = in_array[high]

        out_array.append(a * (1 - weight) + b * weight)
```

```
    return np.array(out_array)

def bilinear_resize(image, height, width):
    """
    Parameters
    ----------
    image : 2D numpy array.
    height : desired height
    width : desired width

    Returns
    -------
    resized image

    """
    img_height, img_width, img_channels = image.shape

    resized = np.zeros((height, width, img_channels))

    x_ratio = float(img_width - 1) / (width - 1) if width > 1 else 0
    y_ratio = float(img_height - 1) / (height - 1) if height > 1 else 0

    for i in range(height):
        for j in range(width):
            x_l, y_l = math.floor(x_ratio * j), math.floor(y_ratio * i)
            x_h, y_h = math.ceil(x_ratio * j), math.ceil(y_ratio * i)

            x_weight = (x_ratio * j) - x_l
            y_weight = (y_ratio * i) - y_l

            a = image[y_l, x_l]
            b = image[y_l, x_h]
            c = image[y_h, x_l]
            d = image[y_h, x_h]

            pixel = a * (1 - x_weight) * (1 - y_weight)\
                + b * x_weight * (1 - y_weight)\
                + c * y_weight * (1 - x_weight)\
                + d * x_weight * y_weight

            resized[i, j] = pixel

    return resized.astype(np.uint8)

def bilinear_resize_vectorized(image, height, width):
    """
    Parameters
    ----------
    image : 2D numpy array.
    height : desired height
    width : desired width

    Returns
    -------
    resized image

    """
    img_height, img_width, img_channels = image.shape
    resized_img = []

    print(img_height)
    print(img_width)
    for c in range(img_channels):
        img_c = image[:, :, c].ravel()

        x_ratio = float(img_width - 1) / (width - 1) if width > 1 else 0
        y_ratio = float(img_height - 1) / (height - 1) if height > 1 else 0

        y, x = np.divmod(np.arange(height * width), width)

        x_l = np.floor(x_ratio * x).astype(np.uint32)
        y_l = np.floor(y_ratio * y).astype(np.uint32)

        x_h = np.ceil(x_ratio * x).astype(np.uint32)
        y_h = np.ceil(y_ratio * y).astype(np.uint32)

        x_weight = (x_ratio * x) - x_l
        y_weight = (y_ratio * y) - y_l

        a = img_c[y_l * img_width + x_l]
        b = img_c[y_l * img_width + x_h]
        c = img_c[y_h * img_width + x_l]
        d = img_c[y_h * img_width + x_h]

        resized = a * (1 - x_weight) * (1 - y_weight) + \
                  b * x_weight * (1 - y_weight) + \
                  c * y_weight * (1 - x_weight) + \
                  d * x_weight * y_weight
        resized_img.append(resized.reshape(height, width).astype(np.uint8))
```

```
    return np.stack(resized_img, axis=2)

def bilinear_resize_vectorized_backward(image, height, width, grad):
    """
    `image` is a 2-D array, holding the input image
    `height` and `width` are the desired spatial dimension of the new 2-D array.
    `grad` is a 2-D array of shape [height, width], holding the gradient to be
    backpropped to `image`.
    """
    # This backprop implementation only works for a single channel
    img_height, img_width, _ = image.shape
    image = image.ravel()
    x_ratio = float(img_width - 1) / (width - 1) if width > 1 else 0
    y_ratio = float(img_height - 1) / (height - 1) if height > 1 else 0

    y, x = np.divmod(np.arange(height * width), width)

    x_l = np.floor(x_ratio * x).astype('int32')
    y_l = np.floor(y_ratio * y).astype('int32')

    x_h = np.ceil(x_ratio * x).astype('int32')
    y_h = np.ceil(y_ratio * y).astype('int32')

    x_weight = (x_ratio * x) - x_l
    y_weight = (y_ratio * y) - y_l

    grad = grad.ravel()
    # gradient wrt `a`, `b`, `c`, `d`
    d_a = (1 - x_weight) * (1 - y_weight) * grad
    d_b = x_weight * (1 - y_weight) * grad
    d_c = y_weight * (1 - x_weight) * grad
    d_d = x_weight * y_weight * grad
    # [4 * height * width]
    grad = np.concatenate([d_a, d_b, d_c, d_d])
    # [4 * height * width]
    indices = np.concatenate([y_l * img_width + x_l,
    y_l * img_width + x_h,
    y_h * img_width + x_l,
    y_h * img_width + x_h])
    # we must route gradients in `grad` to the correct indices of `image` in
    # `indices`, e.g. only entries of indices `y_l * img_width + x_l` in `image`
    # gets the gradient backpropped from `a`.
    # use numpy's broadcasting rule to generate 2-D array of shape
    # [4 * height * width, img_height * img_width]
    indices = (indices.reshape((-1, 1)) ==
    np.arange(img_height * img_width).reshape((1, -1)))
    d_image = np.apply_along_axis(lambda col: grad[col].sum(), 0, indices)
    return d_image.reshape((img_height, img_width))


if __name__ == '__main__':
    """ 1D Example """
    in_array = [1,2,3,4]
    out_array_smaller = linear_resize(in_array, 2)
    print(out_array_smaller)
    out_array_larger = linear_resize(in_array, 7)
    print(out_array_larger)

    """ 2D Example """
    # Load image
    img=cv2.imread('bender.png')
    print("Original image size", img.shape)
    cv2.imshow('image', img)
    cv2.waitKey(0)
    cv2.destroyAllWindows()

    # New image size
    height = 128
    width = 400
    # Bilinear resize without vectorize
    resized_img = bilinear_resize(img, height, width)
    print("resized_img.shape", resized_img.shape)
    cv2.imshow('Resized image', resized_img)
    cv2.waitKey(0)
    cv2.destroyAllWindows()

    # Using vectorized implementation (faster)
    resized_img = bilinear_resize_vectorized(img, height, width)
    print("resized_img.shape", resized_img.shape)
    cv2.imshow('Resized image', resized_img)
    cv2.waitKey(0)
    cv2.destroyAllWindows()

    """ Test bilinear resize vectorized backward """
    # Create pseudo tensor
    input_tensor = torch.randn(1, 3, 5, 8)  # (N, C, H, W)

    ### Pytorch implementation
    input_tensor.requires_grad_(True)
    # Forward pass
```

```
    output_tensor = F.interpolate(input_tensor, size=(6, 9), align_corners=True, mode='bilinear')
    # Backward pass
    # output_tensor.sum().backward()
    output_tensor.sum().backward()
    torch_grad = input_tensor.grad
    print("torch_grad", torch_grad)
    print("torch_grad.shape", torch_grad.shape)

    ### Manual implementation
    print("")
    # Convert (N, C, H, W) to (H, W, C) since our manual backprop only works for a single image
    print("input_tensor.squeeze(0, 1)[:, :, np.newaxis].shape", input_tensor.squeeze(0).permute(1, 2, 0).shape)
    grad = torch.ones((6, 9))  # Pytorch does this internally during backward
    manual_grad = bilinear_resize_vectorized_backward(input_tensor.squeeze(0).permute(1, 2, 0).detach().cpu().numpy(),
                                                      6, 9, grad.detach().cpu().numpy())
    print("manual_grad", manual_grad)
    print("manual_grad.shape", manual_grad.shape)
```

Output:

```
[1. 4.]
[1.  1.5 2.  2.5 3.  3.5 4. ]
Original image size (338, 600, 3)
resized_img.shape (128, 400, 3)
338
600
resized_img.shape (128, 400, 3)
torch_grad tensor([[[[1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500]],

         [[1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500]],

         [[1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500],
          [1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500, 1.3500]]]])
torch_grad.shape torch.Size([1, 3, 5, 8])

input_tensor.squeeze(0, 1)[:, :, np.newaxis].shape torch.Size([5, 8, 3])
manual_grad [[1.35 1.35 1.35 1.35 1.35 1.35 1.35 1.35]
 [1.35 1.35 1.35 1.35 1.35 1.35 1.35 1.35]
 [1.35 1.35 1.35 1.35 1.35 1.35 1.35 1.35]
 [1.35 1.35 1.35 1.35 1.35 1.35 1.35 1.35]
 [1.35 1.35 1.35 1.35 1.35 1.35 1.35 1.35]]
manual_grad.shape (5, 8)
```

Related:

1. Spatial Transformer network
2. Fully Convolutional Network (FCN)

References:

1. https://chao-ji.github.io/jekyll/update/2018/07/19/BilinearResize.html
2. https://kevinzakka.github.io/2017/01/10/stn-part1/